

Out-of-field teaching—An opportunity for cross-curricular teaching in a complex world?—A scoping review

Romana Dorfer  · James Loparics  · Bettina Krepper · Laura Seiler

Accepted: 15 July 2026
© The Author(s) 2026

Abstract Out-of-field teaching (OOFT)—defined as the practice of teachers teaching subjects outside their formal training, excluding lateral entrants—is a growing phenomenon driven by global challenges such as teacher shortages and large-scale attritions. This scoping review maps the current state of research by analysing 54 studies retrieved from four academic databases, supplemented by expert feedback. The review aims to describe what the literature reports about OOFT: how it is conceptualised; which outcomes are reported (beneficial, neutral, and adverse) and how they are operationalised; which types of evidence and study designs are used; and which supports are documented at teacher, school, and system levels. Findings indicate a dual profile. Risks include gaps in content and pedagogical content knowledge, professional strain, and slower development of high-quality practice for novices; opportunities include teacher learning, identity development, and the adoption of innovative, student-centred pedagogies under supportive conditions. Reported supports span job-embedded professional learning (mentoring, co-planning/co-teaching, peer observation), school-level arrangements (time for collaboration, curated materials, assessment moderation, scheduling that limits OOFT preparations), and longitudinal qualification programmes tailored to prior knowledge. Distributional patterns show that OOFT is more prevalent in disadvantaged and rural contexts, underscoring equity concerns. The review identifies gaps regarding school-based support structures (e.g., mentoring, subject-specific collaboration) and the role of cross-curricular competencies emphasised in many curricula. Overall,

Use of AI The language of this manuscript was edited and formatted using artificial-intelligence tools. All scientific content, analyses and conclusions were independently developed by the authors.

✉ Romana Dorfer · James Loparics · Bettina Krepper · Laura Seiler
Johannes Kepler University, Linz, Austria
E-Mail: romana.dorfer@jku.at

the review offers a map of concepts, outcomes, evidence types, and supports related to OOF and highlights priorities for future research and policy.

Keywords Out-of-field teaching · Cross-curricular work · Teacher training · Post-qualification

Fachfremdes Unterrichten – Eine Chance für fächerübergreifendes Unterrichten in einer komplexen Welt? – Ein Scoping Review

Zusammenfassung Fachfremdes Unterrichten – verstanden als das Unterrichten von Fächern außerhalb der formalen Lehramtsausbildung und ohne Einbezug von Quereinsteigenden – nimmt angesichts globaler Herausforderungen wie Lehrkräftemangel und Pensionierungswellen zu. Dieses Scoping Review kartiert den aktuellen Forschungsstand durch die Analyse von 54 Studien aus vier wissenschaftlichen Datenbanken, ergänzt durch Expert:innenfeedback. Ziel ist es, zu beschreiben, was die Literatur über fachfremdes Unterrichten (FFU) berichtet: wie FFU konzeptualisiert wird, welche Outcomes (vorteilhaft, neutral, unerwünscht) berichtet und wie sie operationalisiert werden, welche Evidenzarten und Studiendesigns vorliegen sowie welche Unterstützungsmaßnahmen auf Lehrkraft-, Schul- und Systemebene dokumentiert sind. Die Befunde zeigen ein duales Profil. Risiken umfassen Lücken im fachlichen und fachdidaktischen Wissen, berufliche Belastungen und eine langsamere Entwicklung qualitativ hochwertigen Unterrichts bei Berufseinsteiger:innen; Chancen bestehen in berufsbiografischem Lernen, Identitätsentwicklung und der Aneignung innovativer, schülerzentrierter Didaktiken unter unterstützenden Bedingungen. Berichtete Unterstützungen reichen von arbeitsintegriertem Professionalisieren (Mentoring, Co-Planning/Co-Teaching, kollegiale Hospitation) über schulische Arrangements (Zeitfenster für Kooperation, kuratierte Materialien, Assessmentmoderation, Stundenplanung zur Begrenzung fachfremder Vorbereitungen) bis zu mehrjährigen Qualifizierungen, die an Vorwissen anknüpfen. Verteilungsmuster zeigen eine höhere FFU-Prävalenz in benachteiligten und ländlichen Kontexten – mit Gerechtigkeitsimplikationen. Forschungslücken betreffen insbesondere schulische Unterstützungsstrukturen (z.B. Mentoring, fachspezifische Kooperation) und die Rolle fächerübergreifender Kompetenzen. Insgesamt liefert das Review eine Kartierung von Konzepten, Outcomes, Evidenzarten und Unterstützungen zu FFU und leitet Prioritäten für Forschung und Bildungspolitik ab.

Schlüsselwörter Fachfremdes Unterrichten · Fächerübergreifende Arbeit · Lehrkräfteausbildung · Weiterbildungen

1 Introduction

Various situations lead to teachers being assigned to subjects outside their formal training (Hobbs and Porsch 2021; McConney and Price 2009). This may result from local substitutions, structural shortages, generational changes, or the introduction of

new subjects without established training. These challenges have also been amplified by the internationally discussed ‘Great Resignation’: since around 2021, the term has been used to describe elevated quit rates and occupational mobility, alongside shifts towards part-time and flexible work (Formica and Sfodera 2022). In the education sector, these trends have intensified teacher shortages and the need for flexible staffing arrangements (Institut des Bundes für Qualitätssicherung im österreichischen Schulwesen 2021; Schmiedehaus et al. 2023; United Nations 2024), making OOFT more prevalent. While OOFT was initially framed from a deficit perspective, researchers have noted that teachers often adapt and develop their skills in response to these challenges (Du Plessis 2016; Sharplin 2014). This paper explores internationally dominant perspectives on OOFT, identifies measures for action, and considers the spectrum of reported outcomes (beneficial, neutral, and adverse). We adopt a pragmatic framing centred on boundary crossing (Akkerman and Bakker 2011) and opportunity structures—how subject alignment, school supports, and allocation patterns shape risks and enable cross-curricular teaching (Hobbs 2013a, b).

Against this backdrop, we adopt a scoping perspective: rather than synthesising effects, we map how OOFT is defined and operationalised across systems; which outcomes are reported and how they are measured; what kinds of evidence underpin claims; and which supports are documented. This meta-level orientation aligns with our interest in boundary crossing and opportunity structures and accommodates heterogeneous contexts and study designs.

2 Aim of the study

In the wake of staffing pressures, evolving knowledge demands, and a growing emphasis on cross-curricular thinking, this study re-examines OOFT from a mapping perspective. Rather than estimating effects, the review describes how OOFT is conceptualised across systems, what outcomes are reported and how they are measured, what types of evidence underpin claims, and what supports are documented at different levels. Consistent with boundary-crossing perspectives, we attend to both risks and opportunities and to the conditions under which each is observed.

This scoping review is guided by the following research questions:

RQ1 (primary) What effects and outcomes of out-of-field teaching are reported in the literature across teacher, classroom/student, school, and system levels, and how are these outcomes defined and measured?

- RQ1a: How is out-of-field teaching conceptualised and operationalised (definitions, subject proximity, educational level)?
- RQ1b: Which outcome domains (beneficial, neutral, adverse) are reported, and which instruments/indicators and time frames are used?

RQ2 (meta-evidence perspective) What types of evidence and claims underpin the literature on out-of-field teaching, and what supports are described?

- RQ2a: What study designs and evidence sources are represented (empirical vs. non-empirical; qualitative, quantitative, mixed; longitudinal, pre-post, reviews)?
- RQ2b: What teacher-, school-, and system-level support measures (e.g., mentoring, collaboration, qualification programmes) are reported, and how are they linked to reported outcomes?

To ensure conceptual coherence, the study excludes lateral entrants without formal teacher education, focusing on qualified in-service teachers assigned outside their field. Given cross-national variation in qualification standards, we analyse opportunity structures (e.g., cross-curricular approaches, professional development pathways) rather than harmonising certification regimes.

Although widely used, the definition of non-specialist teaching remains context dependent. Commonly, OOFT refers to teachers assigned to subjects outside their formal area of training (Hobbs and Porsch 2021; McConney and Price 2009). In German-speaking contexts, it is defined by the absence of formal subject-specific qualifications (Bosse and Törner 2017; Porsch and Gräsel 2022). Building on prior work, we structure OOFT along two dimensions:

- Subject-specific qualification: Porsch (2016) distinguishes between autodidacts (no formal training in the assigned subject), semi-professionals (partial training), and experts (complete training).
- Educational level: OOFT can also occur across educational levels, for example, secondary-trained teachers assigned to primary education (McConney and Price 2009).

Within this framework, *interdisciplinary graduates* represent classificatory edge cases. Their formal degrees integrate multiple disciplines rather than single-subject specialisation, meaning they may hold substantial assignment-relevant expertise without a formal qualification in the specific subject taught.

Cross-national variation in teacher qualification requirements presents a central challenge for this scoping review. Rather than seeking to harmonise divergent national standards, the study adopts an analytic focus on opportunity structures, including cross-curricular teaching approaches and pathways of professional development, both formal and informal, through which subject-specific and pedagogical competencies may be developed (Hobbs 2013a, b; Hobbs and Törner 2019; Porsch and Whannell 2019).

3 Methodology

To provide an overview of the research field and related trends in OOFT, a scoping review was conducted. This approach, which emphasises exploratory and qualitative studies (von Elm et al. 2019), is distinguished from systematic reviews by its broader focus. The review adhered to the five-stage framework by Arksey and O'Malley (2005), refined by Levac et al. (2010), ensuring transparency and replicability. The stages included (1) identifying research questions, (2) identifying studies, (3) selecting studies, (4) charting data, and (5) summarising results.

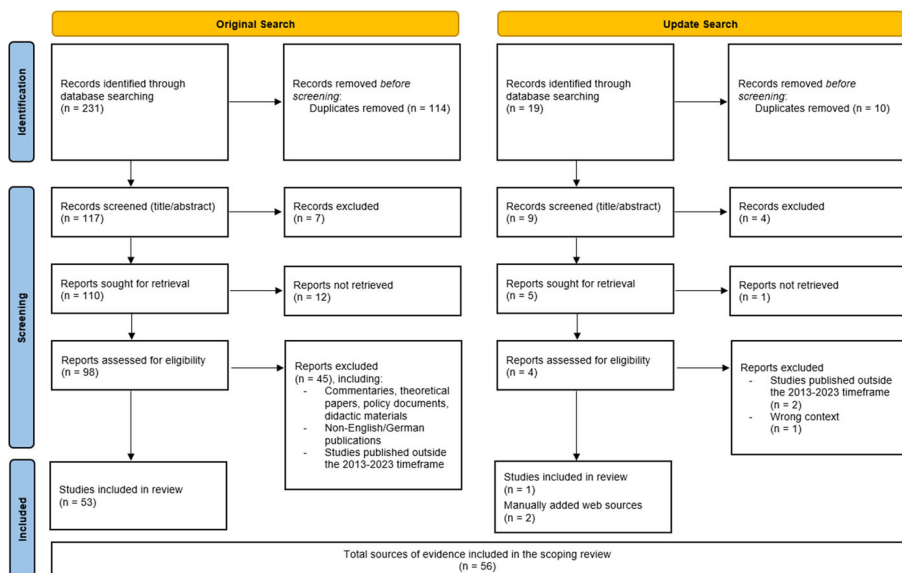


Fig. 1 PRISMA diagram, identified studies and inclusion/exclusion

The study was conducted between December 2022 and June 2023 and was methodologically revised in 2025 based on external expert feedback following a scholarly exchange in 2024. Refinements included targeted adjustments to the search strategy (e.g., through the addition of keywords). Summary yields are reported in the PRISMA diagram (Fig. 1), and full, database-specific search strings are provided in the appendix. Literature searches were conducted in four databases: Web of Science, Scopus, ERIC, and the German-language database ‘fachportal-paedagogik.de’. The English-language databases were searched for the term ‘out-of-field-teaching’, supplemented, based on the 2024 expert feedback, by ‘teaching-across-specialisation’ and ‘non-specialist-teaching’. The German-language database for ‘fachfremder Unterricht’ and in addition (added in 2025) for ‘Unterrichten-ohne-Lehrbefähigung’.

Inclusion criteria focused on qualitative and quantitative studies reporting effects on performance, processes, or actors. Both peer-reviewed journal articles and relevant grey literature, including doctoral dissertations, were eligible. Grey literature was included to reduce publication bias and capture a broader range of empirical evidence relevant to the review questions. Studies published between 2013 and 2023 were included to capture contemporary developments in the field. Only English-language and German-language publications were considered due to feasibility constraints. Excluded were commentaries, theoretical or conceptual papers, policy documents, and didactic or instructional materials, as well as studies outside the defined timeframe or language scope. The search strings can be found in the appendix. We developed the evaluation categories iteratively, drawing on the review aims and patterns observed during screening and full text analysis based on recurring themes,

Table 1 Databases, included studies and additional sources

	Studies actually included in analysis after reading the full text	Studies actually included in analysis after reading the full text
	Original Search	Update Search
Fachportal Pädagogik	10	–
<i>Original Search</i>	(Binner 2021)	
‘fachfremd’	(Bosse and Törner 2017)	
29.12.2022	(Eichholz 2018)	
<i>Update Search</i>	(Hammel 2012)	
	(Koch et al. 2015)	
‘Unterrichten-ohne-Lehrbefähigung’	(Lagies 2020)	
06.03.2025	(Lehmann-Wermser and Weishaupt 2020)	
	(Porsch 2015)	
	(Schiemann 2016)	
	(Ziegler and Richter 2017)	
ERIC	18	1
<i>Original Search</i>	(Cinkir and Kurum 2015)	
‘Out-of-field teaching’	(Donitsa-Schmidt et al. 2021)	
29.12.2022	(Du Plessis et al. 2015)	
<i>Update Search</i>	(Du Plessis 2020)	
	(Goos and Guerin 2021)	
‘teaching-across-specialisation’; ‘non-specialist-teaching’	(Goos et al. 2019)	
06.03.2025	(Hanuscin et al. 2020)	
	(Hobbs 2020)	
	(Napier et al. 2020)	(Crisan and Rodd 2017)
	(Nixon et al. 2017)	
	(Pacana et al. 2019)	
	(Regional Educational Laboratory Northeast and Islands 2020)	
	(Sharplin 2014)	
	(Sheppard et al. 2020)	
	(Taylor et al. 2020)	
	(Vale et al. 2021)	
	(Weldon 2016)	
	(Zhou 2014)	

methodological approaches, and observed patterns, ensuring a comprehensive and meaningful classification of included studies.

The selection process, detailed in Fig. 1, identified 250 studies, with 124 duplicates removed. Title and abstract screening resulted in 115 reports assessed for availability, with 102 full texts successfully retrieved and evaluated for eligibility. Ultimately, 54 publications met the inclusion criteria and were included in the review.

Table 1 (Continued)

	Studies actually included in analysis after reading the full text	Studies actually included in analysis after reading the full text
	Original Search	Update Search
Scopus	13	—
<i>Original Search</i>	(Du Plessis and McDonagh 2021)	
‘Out-of-field teaching’	(Du Plessis 2019)	
09.02.2023	(Lane and Ní Ríordáin 2020)	
<i>Update Search</i>	(Ní Ríordáin et al. 2017)	
	(O’Meara and Faulkner 2022)	
‘teaching-across-specialisation’; ‘non-specialist-teaching’	(Qin and Bowen 2019)	
06.03.2025	(Sani and Burghes 2022)	
	(Schlendorf et al. 2022)	
	(Singh et al. 2021)	
	(Vale et al. 2020)	
	(Du Plessis et al. 2014)	
	(Hobbs and Quinn 2021)	
Web of Science	12	—
<i>Original Search</i>	(Fitchett et al. 2021)	
‘Out-of-field teaching’	(Donitsa-Schmidt and Zuzovsky 2016)	
02.03.2023	(Du Plessis 2015)	
<i>Update Search</i>	(Du Plessis et al. 2014)	
	(Du Plessis 2016)	
‘teaching-across-specialisation’; ‘non-specialist-teaching’	(Du Plessis 2018)	
06.03.2025	(Goos et al. 2021)	
	(Hill and Dalton 2013)	
	(Hobbs 2013b)	
	(Huo et al. 2021)	
	(Shah et al. 2019)	
	(Wittmann et al. 2015)	
Additional sources	—	Teaching out-of-field—network for researchers from German-speaking countries: (PsySensus) https://www.psysensus.de/ Out-of-Field Teaching Across Specialisations OOFTAS Collective: (OOFTAS Collective) https://ooftas-collective.org/

In addition to database searches, a small number of relevant web-based sources were identified through targeted manual web searches conducted during the update phase. These searches focused on established literature hubs and professional networks in the field and applied the same core keywords as the database search.

Consistent with scoping review methodology (Arksey and O'Malley 2005; Levac et al. 2010), we characterise the evidence base (study designs and evidence types; empirical vs. non-empirical claims) and map reported outcomes and supports. We did not conduct a formal risk-of-bias or effect-size synthesis. Data-charting variables were derived from the research questions (conceptualisation/operationalisation of OOFT; population/setting; study design; outcome domains and measures; direction of reported outcomes; support measures).

Table 1 summarises the studies identified in the initial search (2022/23) and the 2025 update. As previously stated, the updated phase was followed by expert feedback, which led to an expansion of the search terms. To enhance transparency and comparability, we compiled a comprehensive tabular overview of all 54 studies (country, method/design, levels of analysis, main findings) provided in the appendix.

4 Research findings

This review synthesises international scholarship on OOFT with two aims: to assess reported effects and to identify effective supports. The corpus comprises 23 qualitative studies (including interviews, classroom observations, and document analyses), 21 quantitative studies, and 8 mixed-methods studies. Across these, 6 adopt longitudinal designs and 2 employ pre-post assessments. A study-level synopsis is available in Appendix Table 3. Two web-based sources were additionally consulted as literature and networking hubs: one primarily addressing German-speaking researchers (<https://www.psysisensus.de/>) and one with an international focus (<https://oofas-collective.org>). Across contexts, evidence is heterogeneous and often correlational. Findings are shaped by allocation policies, school location, and socio-economic factors (Sheppard et al. 2020; Ziegler and Richter 2017; Qin and Bowen 2019). Programme evaluations vary in duration and design rigour, with few long-term follow-ups (Donitsa-Schmidt et al. 2021; Sani and Burghes 2022). These limitations inform the interpretations offered below.

4.1 Effects of out-of-Field teaching

4.1.1 Teacher-Level effects

Studies consistently document risks for teachers assigned outside their specialisation, including gaps in content and pedagogical content knowledge (Ní Ríordáin et al. 2017; Cinkir and Kurum 2015), diminished motivation and job satisfaction (Sharplin 2014), persistent professional insecurities and shortcomings even after several years (Wittmann et al. 2015), fatigue and self-doubt (Fitchett et al. 2021; Du Plessis et al. 2014; Hammel 2012), and higher attrition risks (Donitsa-Schmidt and Zuzovsky 2016). These patterns are particularly pronounced in mathematics and science (Ní

Ríordáin et al. 2017; Sharplin 2014). Early-career teachers face specific challenges, including isolation and weaker subject-specific pedagogies (Nixon et al. 2017; Singh et al. 2021).

At the same time, OOFT can open development trajectories: it may broaden employment prospects and perspectives on teaching (Cinkir and Kurum 2015), support professional identity work (Crisan and Rodd 2017), and shift beliefs towards more constructivist orientations (Vale et al. 2020). Teachers with adaptive, improvement-focused dispositions engage more productively with OOFT demands such as working with unfamiliar content, differentiating instruction, aligning assessment with curricular goals, and navigating organisational constraints (Lagies 2020). These demands cluster into content, pedagogy, resources, and organisational routines and are best addressed through job-embedded supports (mentoring, peer observation, co-planning/co-teaching, lesson study). Variation in effects is moderated by motivational orientation (extrinsic vs. intrinsic; Donitsa-Schmidt et al. 2021), career stage (novice vs. experienced; Hanuscin et al. 2020), and the proximity of the assigned subject to a teacher's prior qualification (Koch et al. 2015). In this context, *integrative subject background* denotes degree programmes that combine related disciplinary domains (e.g., geography-economics, general science, social studies), which afford structural overlap with the assigned subject and can ease knowledge transfer.

4.1.2 Classroom practice and student learning

Evidence on classroom practice suggests that, under supportive conditions, OOFT can be associated with more student-centred, problem-based, and application-oriented approaches. Out-of-field teachers often adopt training-derived models intensively (Goos and Guerin 2021), and sustained OOFT experience has been linked to shifts from procedural instruction towards problem-solving and deeper understanding (Vale et al. 2021; O'Meara and Faulkner 2022).

Findings on student achievement are mixed: several studies report lower outcomes in OOFT contexts, yet these differences diminish substantially once socio-economic and allocation factors are controlled (Sheppard et al. 2020; Ziegler and Richter 2017). Longitudinal comparisons indicate that novices in OOFT typically need more time than in-field counterparts to establish high-quality practices (Napier et al. 2020).

4.1.3 Distributional and system-Level patterns

OOFT is disproportionately concentrated in disadvantaged and rural schools across several systems (Regional Educational Laboratory Northeast and Islands 2020; Schlendorf et al. 2022; Taylor et al. 2020; Sharplin 2014; Weldon 2016; Ziegler and Richter 2017). Government allocation policies shape the distribution of OOFT positions (Qin and Bowen 2019), and there is concern that OOFT can reinforce inequity when more qualified teachers are systematically assigned to higher-performing student populations (Hill and Dalton 2013). Evidence on the role of school-level management characteristics is mixed, though some cross-national analyses suggest that greater autonomy in staffing is associated with lower OOFT prevalence (Zhou 2014).

4.1.4 Synthesis

OOFT entails a dual profile. Risks include content misalignment, professional strain, and slower development of high-quality practice for novices (Napier et al. 2020). Opportunities include teacher learning, identity development, and the adoption of innovative pedagogies under supportive conditions (Crisan and Rodd 2017; Goos and Guerin 2021; Vale et al. 2021). The net balance depends on moderators across levels: at the teacher level (prior knowledge, motivation; Donitsa-Schmidt et al. 2021; Koch et al. 2015), the classroom and school level (mentoring, collaboration, access to resources; Goos and Guerin 2021), and the system level (allocation policies, resource distribution; Qin and Bowen 2019; Ziegler and Richter 2017). These patterns motivate the support frameworks outlined in Section 4.2.

4.2 Supporting out-of-Field teaching: strategies and structures

4.2.1 Teacher-Initiated coping strategies

OOFT places demands on five interrelated domains: subject content, instructional strategies, conceptions of student learning, the school environment, and the professional self (Hobbs 2020). Novice and experienced teachers display distinct needs and benefit from tailored induction that builds on prior knowledge (Hanuscin et al. 2020). Didactic models for non-specialists emphasise engaging openings, didactic reduction, cooperative learning, and real-world application (Pacana et al. 2019). Teachers vary in their focus (subject mastery vs. organisational routines) and in their adaptive stance (Lagies 2020). Priorities also differ between non-specialist and specialist teachers (e.g., collaboration vs. time management; Porsch 2015). Transitions between out-of-field and in-field assignments are fluid and not inherently problematic (Hobbs, 2013a, b). Individual strategies are most effective when embedded in job-embedded professional learning, mentoring, peer observation, co-planning and co-teaching, and lesson study, which together build capacity for adaptation (Goos and Guerin 2021; Lane and Ní Ríordáin 2020).

4.2.2 School-Level support structures

School leadership and organisational climate are central to how OOFT is experienced and enacted. Leadership that neglects wellbeing and professional support correlates with trust deficits, lower self-confidence, and weaker school attachment. (Du Plessis et al. 2014; Du Plessis 2019; Du Plessis and McDonagh 2021). Across these studies, leadership patterns associated with trust and confidence issues are characterised as managerialist or authoritarian (compliance-focused, deficit framed feedback, limited instructional guidance). In contrast, instructional and transformational leadership, marked by clear pedagogical vision, developmental feedback, and shared responsibility, correlates with more positive experiences for out-of-field teachers. Conversely, coherent frameworks and access to resources can sustain high teaching quality even under OOFT assignments (Du Plessis 2016; Du Plessis et al. 2014). Effective organisational features include clear role expectations, protected

time for collaboration, and curated access to subject-specific instructional materials (Bosse and Törner 2017). In operational terms, effective support systems include (i) formal mentoring and induction tailored to OOFT assignments; (ii) protected time for co-planning, observation, and joint reflection cycles; (iii) curated repositories of syllabi, exemplar units, and assessment rubrics for the target subject; (iv) assessment moderation within subject teams; (v) timetabling that minimises the number of out-of-field preparations; (vi) reduced teaching load during the initial OOFT phase; and (vii) access to subject-specific coaching or networks. Subject specificity matters, particularly in STEM, where practice-oriented support and infrastructure (e.g., lab access, equipment, safety protocols, and subject coaches) are critical (Du Plessis 2020). Cross-national analyses indicate that greater autonomy in staffing may help reduce OOFT incidence (Zhou 2014). While the evidence is regionally concentrated in a small set of systems, converging findings support institutionalised structures for mentoring and collaboration at the school level (Du Plessis 2018; 2019; 2020).

4.2.3 System-Level qualification programmes

Multi-year qualification programmes consistently improve subject knowledge, pedagogical skills, and self-efficacy among out-of-field teachers (Donitsa-Schmidt et al. 2021). In mathematics, targeted programmes have been associated with action-oriented, student-centred, and problem-solving approaches (Goos and Guerin 2021; Goos et al. 2019; Lane and Ní Ríordáin 2020; O'Meara and Faulkner 2022). Shorter, one-year formats yield competence gains, yet sustained impact depends on continued support and collaborative professional learning (Sani and Burghes 2022). Programme effectiveness is strengthened when design features explicitly link exemplar teaching to authentic practice opportunities and independent learning (Eichholz 2018) and when content is tailored to participants' prior knowledge, participants with relevant backgrounds often benefit more (Binner 2021). Programmes can also shift teacher beliefs towards understanding-oriented pedagogy (Vale et al. 2021; O'Meara and Faulkner 2022). The most effective approaches are longitudinal, practice-based, and closely articulated with school-level supports such as mentoring and professional learning communities (Goos and Guerin 2021; Lane and Ní Ríordáin 2020).

4.2.4 Synthesis

An integrated support architecture combines differentiated teacher-level strategies (Hanuscin et al. 2020; Hobbs 2020), institutionalised school-level structures for collaboration and mentoring—including access to subject-specific resources and, where possible, greater staffing autonomy (Bosse and Törner 2017; Zhou 2014; Du Plessis 2016)—and longitudinal, practice-embedded, system-level programmes tailored to prior knowledge and supported by follow-up (Donitsa-Schmidt et al. 2021; Sani and Burghes 2022; Goos and Guerin 2021).

4.3 Integrative discussion and implications

OOFT is a multi-level, context-sensitive phenomenon rather than a uniform problem. Risks and opportunities co-occur, with positive trajectories most likely when OOFT is framed as a supported professional learning process rather than an ad hoc staffing solution (Goos and Guerin 2021; Lane and Ní Ríordáin 2020). For practice, schools should resource mentoring and collaborative planning, curate access to subject-specific materials, and, where feasible, leverage staffing autonomy to minimise misalignment (Bosse and Törner 2017; Zhou 2014). For policy, allocation mechanisms should reduce inequitable distributions (Qin and Bowen 2019; Ziegler and Richter 2017), and investments should prioritise longitudinal, adaptive programmes tightly coupled with school-based supports (Donitsa-Schmidt et al. 2021; Sani and Burghes 2022). For research, rigorous designs that disentangle allocation from teacher effects, comparative work across subjects, and cross-system replications of school-level interventions are needed to refine causal claims and inform scalable models of support (Sheppard et al. 2020; Napier et al. 2020).

5 Discussion

Building on Section 4's heterogeneous, largely correlational evidence, this discussion clarifies how OOFT generates risks and opportunities across teacher, classroom, school, and system levels. It identifies key mechanisms that moderate effects and derives proportionate, multi-level supports aligned with observed allocation patterns, school contexts, and programme designs. We read these patterns through a pragmatic frame—subject alignment, support/resources, school collaboration, and allocation—rather than a single formal theory.

5.1 Effects and mechanisms

The effects profile is dual and context-dependent. Risks are most visible where assignments misalign with teachers' subject expertise, especially in mathematics and science, leading to gaps in content and pedagogical content knowledge and to diminished confidence, motivation, and job satisfaction (Ní Ríordáin et al. 2017; Cinkir and Kurum 2015; Sharplin 2014). These strains can persist and are associated with fatigue, self-doubt, and elevated attrition risk (Donitsa-Schmidt and Zuzovsky 2016; Du Plessis et al. 2014; Hammel 2012; Fitchett et al. 2021; Skaalvik and Skaalvik 2017; Wittmann et al. 2015). Novices typically take longer than in-field peers to establish high-quality practice, producing early outcome gaps (Napier et al. 2020). At the distributional level, OOFT is disproportionately concentrated in disadvantaged and rural schools, often shaped by allocation policies, raising equity concerns when more qualified teachers are matched to higher-performing populations (Regional Educational Laboratory Northeast and Islands 2020; Schlendorf et al. 2022; Taylor et al. 2020; Sharplin 2014; Weldon 2016; Ziegler and Richter 2017; Hill and Dalton 2013; Qin and Bowen 2019).

Balanced against the risks, credible opportunities emerge under supportive conditions. OOFTE can broaden teachers' perspectives, catalyse professional identity work, and foster adaptive orientations towards new content and pedagogy (Cinkir and Kurum 2015; Crisan and Rodd 2017; Lagies 2020). Sustained experience is associated with shifts from procedural instruction to student-centred, problem-solving, and application-oriented approaches, with robust uptake of training-derived models (Goos and Guerin 2021; Vale et al. 2021; O'Meara and Faulkner 2022). Notably, achievement differences between out-of-field and in-field contexts diminish substantially when socio-economic composition and allocation factors are controlled, indicating that part of the observed gaps reflects contextual covariates rather than assignment effects per se (Sheppard et al. 2020; Ziegler and Richter 2017).

Variation in outcomes is patterned by identifiable mechanisms. At the teacher level, motivational orientation, career stage, and the proximity of the assigned subject to prior qualifications shape engagement with unfamiliar content and pedagogies (Donitsa-Schmidt et al. 2021; Hanuscin et al. 2020; Koch et al. 2015). Integrative disciplinary backgrounds (e.g., geography-economics, general science, social studies) create structural overlap and ease knowledge transfer. At the school level, job-embedded supports—mentoring, structured peer observation, co-planning and co-teaching, and lesson study—help address demands across content, pedagogy, resources, and organisational routines (Goos and Guerin 2021; Lane and Ní Ríordáin 2020; Bosse and Törner 2017). Leadership orientation is pivotal: instructional/transformational leadership with clear pedagogical vision, development feedback, and shared responsibility correlates with higher trust and confidence among out-of-field teachers, whereas managerialist approaches are associated with deficit framing and weaker school attachment (Du Plessis et al. 2014; Du Plessis 2019; Du Plessis and McDonagh 2021; Leithwood et al. 2008). At the system level, allocation policies and staffing autonomy influence where and how intensively OOFTE occurs. Greater autonomy can be associated with lower prevalence, whereas policy-driven placements without commensurate supports tend to concentrate risk (Zhou 2014; Qin and Bowen 2019; Ziegler and Richter 2017).

5.2 Support measures

Support measures are most effective when they explicitly target these risks and are bundled across levels. At the teacher level, differentiated induction and role-specific preparation address motivational and knowledge-related variation (Hanuscin et al. 2020; Hobbs 2020). Non-specialist didactics, engaging openings, didactic reduction, cooperative learning, and real-world application, help mitigate content unfamiliarity while maintaining cognitive demand (Pacana et al. 2019). Job-embedded professional learning (mentoring, observation, and co-planning/co-teaching) accelerates the shift from procedural to understanding-oriented instruction and reduces preparation load (Goos and Guerin 2021; Lane and Ní Ríordáin 2020).

At the school level, effective arrangements include protected time for collaboration and observation, curated subject-specific repositories (syllabi, exemplar units, rubrics), assessment moderation, timetabling that minimises the number of out-of-field preparations, reduced teaching load in the initial phase, and access to sub-

ject-specific coaching or networks (Bosse and Törner 2017; Du Plessis 2016; Du Plessis et al. 2014). These measures map onto identified risks: time and reduced load address fatigue; repositories and moderation improve alignment and assessment validity; subject coaching tackles domain-specific hurdles. In STEM, ensuring lab access, equipment, safety protocols, and specialised coaching is essential for practice-oriented instruction (Du Plessis 2020). Leadership that foregrounds instructional guidance and development feedback is a necessary enabling condition for coherent enactment (Du Plessis 2019; Du Plessis and McDonagh 2021).

At the system level, longitudinal, practice-based qualification programmes that integrate subject knowledge, pedagogy, and authentic practice improve competence and self-efficacy. Shorter formats often require continued support to sustain impact (Donitsa-Schmidt et al. 2021; Sani and Burghes 2022, Eichholz 2018). Designs that diagnose prior knowledge and tailor pathways are more effective. Participants with related backgrounds often benefit more (Binner 2021). In mathematics and science, such programmes reliably promote understanding-oriented pedagogy and problem-solving. Coupling programme participation with in-school mentoring and professional learning communities enhances transfer and durability (Goos and Guerin 2021; Lane and Ní Ríordáin 2020; Vale et al. 2021; O'Meara and Faulkner 2022). In parallel, allocation mechanisms and calibrated staffing autonomy should avoid concentrating OOFT in disadvantaged schools without compensatory resource packages and coaching (Qin and Bowen 2019; Ziegler and Richter 2017; Zhou 2014).

5.3 Implications and limitations

Taken together, these patterns support a reframing of OOFT from an ad hoc staffing response to a supported professional learning trajectory. When assignments are proximate to existing expertise, embedded in collaborative school environments, and accompanied by longitudinal, practice-embedded development, positive effects—teacher learning, identity development, and pedagogical innovation—are more likely. Conversely, concentration in high-need contexts without adequate resources and compliance-oriented leadership and policy increases the risks of content misalignment, professional strain, and slower instructional improvement.

The evidence base warrants cautious interpretation and clear statements on generalisability. The corpus is heterogenous and predominantly correlational, with few long-term follow-ups, which limits causal inference and external validity (Sheppard et al. 2020; Ziegler and Richter 2017; Donitsa-Schmidt et al. 2021; Sani and Burghes 2022). The estimation of assignment effects is complicated by confounding factors including allocation, school location and socio-economic composition. Methodological limitations include potential bias related to database selection, language restrictions, and aspects of the search strategy, which may have resulted in the exclusion of some grey or localised literature. Future research should leverage quasi-experimental and longitudinal designs, exploit natural experiments in staffing, and apply propensity score or instrumental-variable strategies where feasible (Sheppard et al. 2020; Napier et al. 2020). Subject-comparative work (STEM vs. humanities) can clarify domain-specific demands and the value of integrative disciplinary backgrounds. Cross-system replications of school-level interventions are needed to

establish scalability, boundary conditions, and cost-effectiveness. For the German-speaking context, differences in governance (federal responsibilities, school autonomy, teacher education pathways) should be considered when assessing transferability and designing supports.

6 Appendix

Table 2 Search Strings

Database	Search string
ERIC	(TI “out-of-field teaching” OR AB “out-of-field teaching” OR TI “out of field teaching” OR AB “out of field teaching” OR TI (“out-of-field” N3 teach*) OR (“out of field” N3 teach*)) OR AB (“out-of-field” N3 teach*) OR (“out of field” N3 teach*)) <i>Limiters</i> Published Date: 2013-01-01 to 2023-12-31 Language: English <i>Update Search</i> (TI “teaching-across-specialisation” OR AB “teaching-across-specialisation” OR TI “teaching across specialisation” OR AB “teaching across specialisation” OR TI “teaching-across-specialization” OR AB “teaching-across-specialization” OR TI “teaching across specialization” OR AB “teaching across specialization” OR (TI (teach* N3 across) AND TI (across N3 speciali?ation*)) OR (AB (teach* N3 across) AND AB (across N3 speciali?ation*))) (TI “non-specialist-teaching” OR AB “non-specialist-teaching” OR TI “non-specialist teaching” OR AB “non-specialist teaching” OR TI “non specialist teaching” OR AB “non specialist teaching” OR TI “nonspecialist teaching” OR AB “nonspecialist teaching” OR TI (“non-specialist” N3 teach*) OR AB (“non-specialist” N3 teach*) OR TI (“non specialist” N3 teach*) OR AB (“non specialist” N3 teach*) OR TI (nonspecialist N3 teach*) OR AB (nonspecialist N3 teach*)) <i>Limiters</i> Published Date: 2013-01-01 to 2023-12-31 Language: English

Table 2 (Continued)

Database	Search string
Web of Science	TS=(“out-of-field teaching” OR “out of field teaching” OR (“out-of-field” NEAR/3 teach*) OR (“out of field” NEAR/3 teach*)) <i>Refinements</i> Timespan: 2013–2023 Language: English <i>Update Search</i> TS=(“teaching-across-specialisation” OR “teaching across specialisation” OR “teaching-across-specialization” OR “teaching across specialization” OR ((teach* NEAR/3 across) AND (across NEAR/3 speciali?ation*))) TS=(“non-specialist-teaching” OR “non-specialist teaching” OR “non specialist teaching” OR “nonspecialist teaching” OR (“non-specialist” NEAR/3 teach*) OR (“non specialist” NEAR/3 teach*) OR (nonspecialist NEAR/3 teach*)) <i>Refinements</i> Timespan: 2013–2023 Language: English
SCOPUS	TITLE-ABS-KEY(“out-of-field teaching” OR “out of field teaching” OR (“out-of-field” W/3 teach*) OR (“out of field” W/3 teach*)) AND (PUBYEAR >= 2013 AND PUBYEAR <= 2023) AND (LANGUAGE(english)) <i>Update Search</i> TITLE-ABS-KEY(“teaching-across-specialisation” OR “teaching across specialisation” OR “teaching-across-specialization” OR “teaching across specialization” OR ((teach* W/3 across) AND (across W/3 speciali?ation*))) AND (PUBYEAR >= 2013 AND PUBYEAR <= 2023) AND (LANGUAGE(english)) TITLE-ABS-KEY(“non-specialist-teaching” OR “non-specialist teaching” OR “non specialist teaching” OR “nonspecialist teaching” OR (“non-specialist” W/3 teach*) OR (“non specialist” W/3 teach*) OR (nonspecialist W/3 teach*)) AND (PUBYEAR >= 2013 AND PUBYEAR <= 2023) AND (LANGUAGE(english))
Fachportal Pädagogik	“fachfremder Unterricht” OR (fachfremd* AND unterricht*) <i>Filter</i> Erscheinungsjahr: 2013–2023 Sprache: Deutsch <i>Update Search</i> “Unterrichten ohne Lehrbefähigung” OR “Unterrichten-ohne-Lehrbefähigung” OR (unterrichten* AND ohne AND (Lehrbefähig* OR Lehrbefachig* OR Lehramtsbefähig* OR Lehramtsbefachig* OR Lehrbefug*)) <i>Filter</i> Erscheinungsjahr: 2013–2023 Sprache: Deutsch

The search strings have been reproduced because only the search parameters were noted down, but not the exact search strings

Table 3 Overview of Included Studies ($n = 54$)

Study	Country	Methods/Designs	Levels of analysis	Main findings
Cinkir and Kurum 2015	Turkey	Qualitative descriptive	Teacher, School leadership, Policy	Policy-driven OOF alleviates teacher shortages but compromises instructional quality. Enhanced collaboration between school administration and the Higher Education Council is recommended
Donitsa-Schmidt et al. 2021	Israel	Quantitative	Teacher, Professional development (PD)	Two motivation patterns (intrinsic and mixed) were identified. PD improved subject and pedagogical content knowledge and self-efficacy, yet measures are needed to prevent quality loss due to limited content knowledge
Du Plessis et al. 2015	Australia	Qualitative longitudinal case studies	Teacher, School leadership	OOF novices face trust and self-confidence issues and weak connectedness. School leadership styles are often misunderstood
Du Plessis 2020	Australia	Qualitative (Interviews, observations, document analysis)	Teacher, School leadership	OOF in science requires deep disciplinary and laboratory/field knowledge. School leaders should provide targeted STEM PD and practical support
Goos and Guerin 2021	Ireland	Quantitative observation	Teacher, Classroom	All teacher groups (in-field, out-of-field, up-skilled) scored high on supportive learning environments. Upskilled OOF teachers most strongly adapted the learned pedagogical practices
Goos et al. 2019	Ireland	Quantitative Online-Survey	Teacher, Beliefs	PD participants displayed child-centred, problem-solving, and conceptual orientations aligned with the new national curriculum
Hanuscin et al. 2020	USA	Quantitative descriptive	Teacher, Induction	Experienced teachers teaching out-of-field require tailored induction due to differing practices, demands, and content. Learning needs differ from novices
Hobbs 2020	Australia	Qualitative longitudinal interviews	Teacher, Professional learning	OOF learning is multidimensional (content strategies, student learning, school, professional self). Connecting OOF content to teachers' original fields is key
Napier et al. 2020	USA	Qualitative longitudinal	Teacher, Classroom practice	OOF intensifies challenges in the first year and fosters isolated scientific practices. Further research is warranted
Nixon et al. 2017	USA	Quantitative longitudinal	Teacher, Policy, Induction	Many early-career teachers teach out-of-field, complicating induction. Targeted support and preparation across multiple disciplines are urged

Table 3 (Continued)

Study	Country	Methods/Designs	Levels of analysis	Main findings
Pacana et al. 2019	Philippines	Qualitative descriptive	Teacher, Curriculum design	In social studies OOFI, a pedagogical model was developed to flexibly respond to contextual demands
Regional Educational Laboratory Northeast and Islands 2020	USA	Quantitative descriptive	System, School	OOFI occurs more frequently in schools classified as high-need/critical
Sharplin 2014	Australia	Qualitative Grounded Theory	Teacher, Induction	Rural OOFI assignments exacerbate induction difficulties. Misassignment increases dissatisfaction. Targeted support structures are needed
Sheppard et al. 2020	USA	Quantitative representative (regression analysis)	Student outcomes, Teacher assignment	Students taught by out-of-field teachers underperform relative to those caught by certified peers. Socio-economic status is the dominant predictor
Taylor et al. 2020	USA	Quantitative representative	System, Teacher experience	Formal certification relates to outcomes mainly in Earth Science. Teacher experience and access to materials matter more in Physics and Life Science. OOFI is more prevalent in middle schools and disadvantaged contexts
Vale et al. 2021	Australia	Qualitative longitudinal	Teacher, Beliefs, Practice	Extend OOFI can shift beliefs towards student-centred, problem-solving approaches through boundary crossing processes
Weldon 2016	Australia	Quantitative descriptive	System	OOFI is widespread (e.g., 26% in Years 7–10), with higher rates in remote areas than metropolitan schools
Zhou 2014	OECD (15 countries)	Quantitative	System, School leadership	School leadership style shows no robust association with OOFI after country controls. Greater school autonomy in staffing correlates with lower OOFI rates
Binner 2021	Germany	Mixed methods	Teacher, PD	Content-focused PD yields larger knowledge gains for teachers with higher prior knowledge. PD cannot replace foundational training but can substantially increase knowledge for out-of-field and certified teachers
Bosse and Törner 2017	Germany	Qualitative phenomenological	Teacher, School conditions	Six out-of-field teacher types were identified. OOFI in mathematics is manageable given appropriate school conditions and supports

Table 3 (Continued)

Study	Country	Methods/Designs	Levels of analysis	Main findings
Eichholz 2018	Germany	Mixed methods	Teacher, PD	Long-term, example-based, practice-oriented PD with school support enhances professional competence for OOFFT in primary mathematics
Hammel 2012	Germany	Qualitative descriptive	Teacher, Wellbeing	OOFT in primary music, coupled with high self-imposed standards, is linked to stress and self-perception challenges
Koch et al. 2015	Germany	Quantitative descriptive	Teacher, Self-assessment	Teachers with disciplinary training rate their energy education knowledge higher than out-of-field teachers
Lagtes 2020	Germany	Qualitative Documentary method	Teacher, Professional identity	OOFT enactments span profession- and organisation-oriented logics. In primary schools, a border 'primary school habitus' may supersede a specific OOFFT habitus
Lehmann-Wermser and Weishaupt 2020	Germany	Quantitative descriptive	System, Policy	Persistent music teacher shortages prompt discussion of timetable reductions, standard lowering, partnerships with music schools, and systematic retraining
Porsch 2015	Germany	Qualitative representative	Teacher, Beliefs	Certified teachers endorse stronger constructivist principles. Out-of-field teachers emphasise collaboration and value training components differently
Schiemann 2016	Germany	Explorative (Videoography)	Teacher, Classroom practice	Marked differences in pedagogical action patterns appear between a trained and an out-of-field music teacher. Findings are exploratory and limited in generalisability
Ziegler and Richter 2017	Germany	Quantitative representative	Inequality, Student outcomes	OOFT is more common in classes with higher migration shares and lower average performance. Teacher certification effects diminish after controlling composition
Du Plessis and McDonagh 2021	Australia/South Africa	Qualitative deep understanding	Teacher, School leadership	Out-of-field teachers' lived experiences with leadership, emotions, and meanings highlight needs for context-sensitive support
Du Plessis 2019	Australia/South Africa	Qualitative deep understanding	Teacher, External factors	Out-of-field teachers' evaluations address performance, content knowledge, self-confidence, communication, and external influences shaping OOFFT

Table 3 (Continued)

Study	Country	Methods/Designs	Levels of analysis	Main findings
Hobbs, 2013b	Australia	Qualitative deep understanding	Teacher, Identity	Teachers report identity and self-conception differences when teaching in-field vs. out-of-field
Lane and Ní Riordáin 2020	Ireland	Qualitative descriptive (Document analysis)	Teacher, Beliefs, Practices	OOFT mathematics teachers documented shifts in beliefs and practices aligned with constructivist orientations
Ní Riordáin et al. 2017	Ireland	Mixed methods	Teacher, Content knowledge	OOFT mathematics teachers showed low performance and frequent conceptual errors in subject content despite high self-reported teaching confidence
O'Meara and Faulkner 2022	Ireland	Mixed Methods	Teacher, PD	Workshops significantly increased perceived mathematics teaching efficacy and shifted self-reported style towards student-centred approaches
Qin and Bowen 2019	OECD (32 countries)	Quantitative	Inequality, School, Classroom distribution	Out-of-field teachers are disproportionately assigned to low SES classes and schools. Within schools' disparities often exceed between-school disparities, varying by country
Sani and Burghes 2022	UK	Qualitative longitudinal	Teacher, PD, Professional support	One-year retraining is insufficient. Sustained collaborative continuing PD is needed to address shortages of competent, confident secondary mathematics teachers
Schendorf et al. 2022	USA	Quantitative	System, Student outcomes	Nearly a quarter of geoscience classes were out-of-field. Student performance was predicted by SES, ethnicity, English proficiency, and instructional exposure
Singh et al. 2021	USA	Mixed methods	Teacher, Induction	Early-career physics teachers teaching out-of-field exhibited less developed and more inconsistent enacted pedagogical content knowledge (PCK). Enacted PCK fluctuated, reflecting its emergent nature
Vale et al. 2020	Australia	Mixed methods	Teacher, Beliefs, Practice transfer	Traditional science views yield similar beliefs across science and mathematics. Constructivist views reduce cross-subject belief coherence
Du Plessis et al. 2014	Australia/South Africa	Mixed methods	School leadership, PD	Out-of-field teachers struggle with self-esteem/confidence and rely on experienced colleagues. Expectations are high while school support is often limited

Table 3 (Continued)

Study	Country	Methods/Designs	Levels of analysis	Main findings
Hobbs and Quinn 2021	Australia	Qualitative deep understanding	Teacher, Emotions, Learning	OOFT settings produced larger gains in perceived ability and enjoyment but more disrupted learning trajectories. Support should be individualised
Fitchett et al. 2021	USA	Quantitative	Teacher, Wellbeing	OOFT is associated with greater susceptibility to workplace fatigue and lower commitment. Health-focused interventions are recommended
Donitsa-Schmidt and Zuzovsky 2016	Israel	Quantitative	System, School leadership	High teacher turnover (including due to OOFT) is documented, along with leadership strategies to manage shortages
Du Plessis 2015	Australia/South Africa	Qualitative deep understanding	Teacher, School leadership	Stakeholder perceptions and experiences of OOFT and instructional quality prompt reconsideration of school improvement and learning environment strategies
Du Plessis et al. 2014	Australia/South Africa	Qualitative deep understanding	Teacher, School leadership	PD shapes OOFT awareness and practices. Effective communication with leadership can reframe OOFT as an opportunity
Du Plessis 2016	Australia/South Africa	Qualitative deep understanding	School leadership, School effectiveness	Informed, engaged principals positively influence leadership approaches, priorities, and supports related to OOFT
Du Plessis 2018	Australia/South Africa	Qualitative deep understanding	Teacher, Diversity, School leadership	OOFT constrains teachers' capacity to address classroom diversity. Leaders should create enabling conditions to meet varied learner needs
Goos et al. 2021	Ireland	Mixed methods	System, Teacher, PD outcomes	PDMT reduced OOFT in mathematics. Graduates reported teaching at higher cognitive levels and more problem-oriented practices
Hill and Dalton 2013	USA	Quantitative	Inequality, Assignment	Out-of-field teachers are disproportionately assigned to disadvantaged schools and not optimally to low-performing students within schools
Hobbs, 2013a	Australia	Qualitative	Teacher support model	A model specifying supports, personal resources, and contextual conditions for effective OOFT was developed from qualitative data

Table 3 (Continued)

Study	Country	Methods/Designs	Levels of analysis	Main findings
Huo et al. 2021	China	Quantitative	System	OOFT is widespread in primary schools. Existing countermeasures are briefly described
Shah et al. 2019	USA	Descriptive (NTPS 2003–2016)	System, Policy	‘No child left behind’-era policies did not reduce OOFT trends in chemistry/physics. OOFT shares and student exposure increased
Wittmann et al. 2015	Germany	Quantitative	Teacher competence	Kindergarten/Primary teachers without mathematics degrees showed insufficient competencies. Long-term OOFT experience did not compensate
Crisan and Rodd 2017	England	Qualitative case studies	Teacher, PD, Professional identity	Reports a research project on designing and implementing preparatory courses for out-of-field mathematics teachers. Examines aspects of mathematics teacher identity in this context

Supplementary Information The online version of this article (<https://doi.org/10.1007/s35834-026-00563-3>) contains supplementary material, which is available to authorized users.

Author Contribution All authors contributed to the conception and design of the study. James Loparics, Bettina Krepper, and Laura Seiler conducted the initial literature search, study selection, data charting, and drafted the first version of the manuscript in year 2023. Romana Dorfer led the further development of the study in year 2025/26, substantially revised the manuscript, conducted an additional phase of data analysis, and prepared the final version, including the final reporting of results and the preparation of figures (e.g., PRISMA diagram). James Loparics contributed to the second phase of analysis and manuscript revision. Bettina Krepper and Laura Seiler reviewed and commented on the manuscript. All authors read and approved the final manuscript.

Funding The author's position is funded by the Austrian Federal Ministry of Education, Science and Research (Bundesministerium für Bildung, Wissenschaft und Forschung; BMBWF) in cooperation with the Innovation Foundation for Education (Innovationsstiftung für Bildung; ISB) and the Austrian Agency for Education and Internationalisation (Österreichs Agentur für Bildung und Internationalisierung; OeAD) through the initiative 'Educational Innovation Needs Educational Research B3', which aims to strengthen the field of educational research by creating cooperative doctoral programmes between universities and teacher training colleges.

Open access funding provided by Johannes Kepler University Linz.

Data Availability Statement Access to the data of the present study can be requested from Romana Dorfer.

Declarations

Conflict of interest R. Dorfer, J. Loparics, B. Krepper and L. Seiler declare that they have no competing interests.

Open Access Dieser Artikel wird unter der Creative Commons Namensnennung 4.0 International Lizenz veröffentlicht, welche die Nutzung, Vervielfältigung, Bearbeitung, Verbreitung und Wiedergabe in jeglichem Medium und Format erlaubt, sofern Sie den/die ursprünglichen Autor(en) und die Quelle ordnungsgemäß nennen, einen Link zur Creative Commons Lizenz beifügen und angeben, ob Änderungen vorgenommen wurden. Die in diesem Artikel enthaltenen Bilder und sonstiges Drittmaterial unterliegen ebenfalls der genannten Creative Commons Lizenz, sofern sich aus der Abbildungslegende nichts anderes ergibt. Sofern das betreffende Material nicht unter der genannten Creative Commons Lizenz steht und die betreffende Handlung nicht nach gesetzlichen Vorschriften erlaubt ist, ist für die oben aufgeführten Weiterverwendungen des Materials die Einwilligung des jeweiligen Rechteinhabers einzuholen. Weitere Details zur Lizenz entnehmen Sie bitte der Lizenzinformation auf <http://creativecommons.org/licenses/by/4.0/deed.de>.

References

- Akkerman, S. F., & Bakker, A. (2011). Learning at the boundary: An introduction. *International Journal of Educational Research*, 50(1), 1–5. <https://doi.org/10.1016/j.ijer.2011.04.002>.
- Arksey, H., & O'Malley, L. (2005). Scoping studies: Towards a methodological framework. *International Journal of Social Research Methodology*, 8(1), 19–32. <https://doi.org/10.1080/1364557032000119616>.
- Binner, E. (2021). *Lernprozesse von qualifikationsheterogenen Grundschullehrkräften im Bereich Stochastik – Studie zur Professionalisierung durch Fortbildung* (Doctoral Dissertation). Humboldt-University Berlin. <https://doi.org/10.18452/22565>
- Bosse, M., & Törner, G. (2017). *Mathematik fachfremd unterrichten: zur Professionalität fachbezogener Lehrer-Identität*. Essener Beiträge zur Mathematikdidaktik. Springer Spektrum.
- Cinkir, S., & Kurum, G. (2015). Discrepancy in teacher employment: The problem of out-of-field teacher employment. *Educational Planning*, 22(1), 29–47.

- Cohen, E., Novis-Deutsch, N., Kashi, S., & Alexander, H. (2024). Interdisciplinary teaching and learning at the K-12 level in the humanities, arts, and social sciences: A scoping review. *Educational Research Review*, 44, 100617. <https://doi.org/10.1016/j.edurev.2024.100617>.
- Crisan, C., & Rodd, M. (2017). Learning mathematics for teaching mathematics: Non-specialist teachers' mathematics teacher identity. *Mathematics Teacher Education and Development*, 19(2), 104–122.
- Donitsa-Schmidt, S., & Zuzovsky, R. (2016). Quantitative and qualitative teacher shortage and the turnover phenomenon. *International Journal of Educational Research*, 77, 83–91. <https://doi.org/10.1016/j.ijer.2016.03.005>.
- Donitsa-Schmidt, S., Zuzovsky, R., & Topaz, B. (2021). Becoming an in-field teacher in Israel: Motivations and knowledge of out-of-field science teachers. *European Journal of Teacher Education*, 44(5), 668–687. <https://doi.org/10.1080/02619768.2020.1860008>.
- Du Plessis, A. E. (2015). Effective education: Conceptualising the meaning of out-of-field teaching practices for teachers, teacher quality and school leaders. *International Journal of Educational Research*, 72, 89–102. <https://doi.org/10.1016/j.ijer.2015.05.005>.
- Du Plessis, A. E. (2016). Leading teachers through the storm: Looking beyond the numbers and turning the implications of out-of-field teaching practices into positive challenges. *International Journal of Educational Research*, 79, 42–51. <https://doi.org/10.1016/j.ijer.2016.06.010>.
- Du Plessis, A. E. (2018). The lived experience of out-of-field STEM teachers: A quandary for strategising quality teaching in STEM? *Research in Science Education*, 50(4), 1465–1499. <https://doi.org/10.1007/s11165-018-9740-9>.
- Du Plessis, A. E. (2019). Barriers to effective management of diversity in classroom contexts: The out-of-field teaching phenomenon. *International Journal of Educational Research*, 93, 136–152. <https://doi.org/10.1016/j.ijer.2018.11.002>.
- Du Plessis, A. E. (2020). *Out-of-field teaching and education policy: International micro-education policy*. <https://doi.org/10.1007/978-981-15-1948-2>.
- Du Plessis, A. E., & McDonagh, K. (2021). The out-of-field phenomenon and leadership for wellbeing: Understanding concerns for teachers, students and education partnerships. *International Journal of Educational Research*. <https://doi.org/10.1016/j.ijer.2020.101724>.
- Du Plessis, A. E., Gillies, R. M., & Carroll, A. (2014). Out-of-field teaching and professional development: A transnational investigation across Australia and south africa. *International Journal of Educational Research*, 66, 90–102. <https://doi.org/10.1016/j.ijer.2014.03.002>.
- Du Plessis, A. E., Carroll, A., & Gillies, R. M. (2015). Understanding the lived experiences of novice out-of-field teachers in relation to school leadership practices. *Asia-Pacific Journal of Teacher Education*, 43(1), 4–21. <https://doi.org/10.1080/1359866X.2014.937393>.
- Eichholz, L. (2018). *Mathematik fachfremd unterrichten*. Wiesbaden: Springer. <https://doi.org/10.1007/978-3-658-19896-1>.
- von Elm, E., Schreiber, G., & Haupt, C. C. (2019). Methodische Anleitung für Scoping Reviews (JBI-Methodologie). *Zeitschrift für Evidenz, Fortbildung und Qualität im Gesundheitswesen*, 143, 1–7. <https://doi.org/10.1016/j.zefq.2019.05.004>.
- Fitchett, P. G., McCarthy, C. J., Lambert, R. G., Eyal, M., Playfair, E. C., & Dillard, J. B. (2021). Examining teacher stress-vulnerability in the US secondary school context. *Educational Review*, 73(2), 170–193. <https://doi.org/10.1080/00131911.2019.1619521>.
- Formica, S., & Sfodera, F. (2022). The great resignation and quiet quitting paradigm shifts: An overview of current situation and future research directions. *Journal of Hospitality Marketing & Management*, 31(8), 899–907. <https://doi.org/10.1080/19368623.2022.2136601>.
- Goos, M., & Guerin, A. (2021). Insights into the pedagogical practices of out-of-field, in-field, and up-skilled teachers of mathematics. In Y. H. Leong, et al. (Ed.), *Excellence in mathematics education: Foundations and pathways (Proceedings of the 43rd Annual Conference of the Mathematics Education Research Group of Australasia)* (pp. 203–210). MERGA.
- Goos, M., Ní Ríordáin, M., Lane, C., & Faulkner, F. (2019). *Impact of a national professional development programme on the beliefs and practices of out-of-field teachers of mathematics*. Annual Meeting of the Mathematics Education Research Group of Australasia (MERGA). <https://files.eric.ed.gov/fulltext/ED604293.pdf>
- Goos, M., Ní Ríordáin, M., Faulkner, F., & Lane, C. (2021). Impact of a national professional development programme for out-of-field teachers of mathematics in Ireland. *Irish Educational Studies*. <https://doi.org/10.1080/03323315.2021.1964569>.
- Hammel, L. (2012). Sich über Diskrepanzen definieren: Selbstkonzepte fachfremd unterrichtender Musiklehrerinnen und Musiklehrer an Grundschulen. Eine Grounded-Theory-Studie. In J. Knigge, A. Niessen & Arbeitskreis Musikpädagogische Forschung (Eds.), *Musikpädagogische Forschung =*

- Research in music education. Musikpädagogisches Handeln: Begriffe, Erscheinungsformen, politische Dimensionen = Music education: concepts, practices, and political dimensions* (pp. 237–255). Verlag Die Blaue Eule.
- Handtke, K., & Böggeholz, S. (2022). The challenge to link biology, chemistry, and physics: Results of a longitudinal study on self-rated content knowledge. *Education Sciences*. <https://doi.org/10.3390/educsci12120928>.
- Hanuscin, D. L., de Araujo, Z., Cisterna, D., Lipsitz, K., & van Garderen, D. (2020). The re-novicing of elementary teachers in science? Grade level reassignment and teacher PCK. *Journal of Science Teacher Education*, 31, 780–801. <https://doi.org/10.1080/1046560X.2020.1778845>.
- Hill, J. G., & Dalton, B. (2013). Student math achievement and out-of-field teaching. *Educational Researcher*, 42(7), 403–405. <https://doi.org/10.3102/0013189X13505682>.
- Hobbs, L. (2013a). Boundary crossings of out-of-field teachers: Location learning possibilities amid disruption. In J. Langan-Fox & C. L. Cooper (Eds.), *Boundary-spanning in organizations: Network, influence, and conflict* (pp. 7–28). <https://doi.org/10.4324/9780203488058>.
- Hobbs, L. (2013b). Teaching 'out-of-field' as a boundary-crossing event: Factors shaping teacher identity. *International Journal of Science and Mathematics Education*, 11(2), 271–297. <https://doi.org/10.1007/s10763-012-9333-4>.
- Hobbs, L. (2020). Learning to teach science out-of-field: A spatial-temporal experience. *Journal of Science Teacher Education*, 31(7), 725–745. <https://doi.org/10.1080/1046560X.2020.1718315>.
- Hobbs, L., & Porsch, R. (2021). Teaching out-of-field: challenges for teacher education. *European Journal of Teacher Education*, 44(5), 601–610.
- Hobbs, L., & Quinn, F. (2021). Out-of-field teachers as learners: Influences on teacher perceived capacity and enjoyment over time. *European Journal of Teacher Education*, 44(5), 627–651. <https://doi.org/10.1080/02619768.2020.1806230>.
- Hobbs, L., & Törner, G. (2019). *Teaching out-of-field as a phenomenon and research problem*. https://doi.org/10.1007/978-981-13-3366-8_1.
- Huo, M., Zhao, N., Zhao, Y., & Noortgate, W. (2021). Who is teaching in Chinese primary schools? A profile of the primary education workforce in Chinese county areas. *PLoS ONE*. <https://doi.org/10.1371/journal.pone.0245549>.
- Institut des Bundes für Qualitätssicherung im österreichischen Schulwesen (2021). *Daten und Material zu Indikatoren B: Input – Personelle und finanzielle Ressourcen. Anhang zum Nationalen Bildungsbericht Österreich 2021 (Teil 2 – Bildungsindikatoren)*. <https://doi.org/10.17888/NBB2021-2-B-DAT>
- Koch, M., Schnell, C., & Malz, S. (2015). Energiebildung unter ökonomischer Perspektive Einschätzungen von Lehrkräften zu ihrem Kenntnisstand und curricularen Anknüpfungspunkten im Wirtschaftsunterricht. *Zeitschrift für ökonomische Bildung*, 3, 56–74.
- Lagies, J. (2020). *Fachfremdheit zwischen Profession und Organisation: Orientierungsrahmen Mathematik fachfremd unterrichtender Grundschullehrkräfte*. Wiesbaden: Springer. <https://doi.org/10.1007/978-3-658-26632-5>.
- Lane, C., & Ní Ríordáin, M. (2020). Out-of-field mathematics teachers' beliefs and practices: An examination of change and tensions using zone theory. *International Journal of Science and Mathematics Education*, 18(2), 337–355. <https://doi.org/10.1007/s10763-019-09964-5>.
- Lehmann-Wermser, A., & Weishaupt, H. (2020). Zur Situation des Musikunterrichts an den Grundschulen und zum zukünftigen Musiklehrkräftebedarf und -angebot. *DDS – Die Deutsche Schule*, 112(1), 41–61. <https://doi.org/10.31244/dds.2020.01.03>.
- Leithwood, K., Harris, A., & Hopkins, D. (2008). Seven strong claims about successful school leadership. *School Leadership & Management*, 28(1), 27–42. <https://doi.org/10.1080/13632430701800060>.
- Levac, D., Colquhoun, H., & O'Brien, K. (2010). Scoping studies: Advancing the methodology. *Implementation Science: IS*, 5, 69. <https://doi.org/10.1186/1748-5908-5-69>.
- McConney, A., & Price, A. (2009). Teaching out-of-field in Western Australia. *Australian Journal of Teacher Education*. <https://doi.org/10.14221/ajte.2009v34n6.6>.
- Napier, J. B., Luft, J. A., & Singh, H. (2020). In the classrooms of newly hired secondary science teachers: The consequences of teaching in-field or out-of-field. *Journal of Science Teacher Education*, 31(7), 802–820. <https://doi.org/10.1080/1046560X.2020.1800195>.
- Ní Ríordáin, M., Paolucci, C., & O'Dwyer, L. M. (2017). An examination of the professional development needs of out-of-field mathematics teachers. *Teaching and Teacher Education*, 64, 162–174. <https://doi.org/10.1016/j.tate.2017.02.001>.
- Nixon, R. S., Luft, J. A., & Ross, R. J. (2017). Prevalence and predictors of out-of-field teaching in the first five years. *Journal of Research in Science Teaching*, 54(9), 1197–1218. <https://doi.org/10.1002/tea.21402>.

- OECD (2020). *TALIS 2018 results*. Vol. 2. OECD. <https://doi.org/10.1787/19cf08df-en>.
- O'Meara, N., & Faulkner, F. (2022). Professional development for out-of-field post-primary teachers of mathematics: An analysis of the impact of mathematics specific pedagogy training. *Irish Educational Studies*, 41(2), 389–408. <https://doi.org/10.1080/03323315.2021.1899026>.
- OOFAS Collective. *Out-of-field teaching across specialisations*. <https://oofas-collective.org>
- Pacana, N. M. S., Ramos, C. D., Catarata, M. N., & Inocian, R. B. (2019). Out-of-field social studies teaching through sustainable culture-based pedagogy: A Filipino perspective. *International Journal of Education and Practice*, 7(3), 230–241. <https://doi.org/10.18488/journal.61.2019.73.230.241>.
- Porsch, R. (2015). Unterscheiden sich Mathematiklehrkräfte an Grundschulen mit und ohne Fach-Lehrbefähigung hinsichtlich ihrer berufsbezogenen Überzeugungen? Ergebnisse aus TIMSS 2007. *Mathematische Didactica*, 38, 5–36.
- Porsch, R. (2016). Herausforderungen für das Lehrerhandeln. *DDS – Die Deutsche Schule*, 108(1), 9–32.
- Porsch, R., & Gollub, P. (2023). *Professionalisierung von Lehrkräften im Beruf. Stand und Perspektiven der Lehrkräftebildung und Professionsforschung*. Waxmann Verlag GmbH. <https://doi.org/10.31244/9783830997429>.
- Porsch, R., & Gräsel, F. (2022). Fachfremdes Unterrichten und Schulleitungen: Einstellungen und Maßnahmen zur Unterstützung fachfremd tätiger Lehrkräfte. *Zeitschrift für Bildungsforschung*, 12(3), 457–478. <https://doi.org/10.1007/s35834-022-00365-3>.
- Porsch, R., & Whannell, R. (2019). *Out-of-field teaching affecting students and learning: What is known and unknown*. https://doi.org/10.1007/978-981-13-3366-8_7.
- PsySensus *Fachfremdes Unterrichten: Netzwerk von Forscherinnen und Forschern aus den deutschsprachigen Ländern zum fachfremd erteilten Unterricht*. <https://www.psysensus.de>
- Qin, L., & Bowen, D. H. (2019). The distributions of teacher qualification: A cross-national study. *International Journal of Educational Development*. <https://doi.org/10.1016/j.ijedudev.2019.102084>.
- Regional Educational Laboratory Northeast & Islands (2020). *Out-of-field teaching among beginning teachers trained and teaching in Rhode Island*. <https://files.eric.ed.gov/fulltext/ED607631.pdf>
- Sani, N., & Burghes, D. (2022). Longitudinal study of 'retraining' non-maths specialist teachers to become capable, confident teachers of mathematics. *International Journal of Mathematical Education in Science and Technology*, 53(9), 2438–2464. <https://doi.org/10.1080/0020739X.2021.1890248>.
- Schiemann, S. (2016). Ein videographisches Auswertungsverfahren zur Erfassung fachdidaktischer Kompetenzen im unterrichtlichen Handeln von (fachfremden) Musiklehrpersonen an Grundschulen. In L. Oravec & A. Steinbach (Eds.), *Musikwissenschaft, Musikpädagogik in der Blauen Eule. Musiklehrpersonen in der Grundschule: Impulse aus Elementarer und schulischer Musikpädagogik* (pp. 173–195). Verlag Die Blaue Eule.
- Schlenker, C., Kelly, A. M., & Krakehl, R. (2022). Earth Science course availability, teacher and school-level characteristics as predictors of precollege Earth Science performance. *Journal of Geoscience Education*. <https://doi.org/10.1080/10899995.2022.2161772>.
- Schmiedehaus, E., Cordaro, M., Perrotte, J., Stern, M., Dailey, S., & Howard, K. (2023). The great resignation in higher education: An occupational health approach to understanding intentions-to-quit for faculty in higher education. *Teaching and Teacher Education*, 123, 103992. <https://doi.org/10.1016/j.tate.2022.103992>.
- Shah, L., Jannuzzo, C., Hassan, T., Gadidov, B., Ray, H. E., & Rushton, G. T. (2019). Diagnosing the current state of out-of-field teaching in high school science and mathematics. *PLoS ONE*. <https://doi.org/10.1371/journal.pone.0223186>.
- Sharplin, E. D. (2014). Reconceptualising out-of-field teaching: Experiences of rural teachers in Western Australia. *Educational Research*, 56(1), 97–110. <https://doi.org/10.1080/00131881.2013.874160>.
- Sheppard, K., Padwa, L., Kelly, A. M., & Krakehl, R. (2020). Out-of-field teaching in chemistry and physics: An empirical census study. *Journal of Science Teacher Education*, 31(7), 746–767. <https://doi.org/10.1080/1046560X.2019.1702268>.
- Singh, H., Luft, J. A., & Napier, J. B. (2021). The development of ePCK of newly hired in-field and out-of-field teachers during their first three years of teaching. *European Journal of Teacher Education*, 44(5), 611–626. <https://doi.org/10.1080/02619768.2021.1943660>.
- Skaalvik, E. M., & Skaalvik, S. (2017). Motivated for teaching? Associations with school goal structure, teacher self-efficacy, job satisfaction and emotional exhaustion. *Teaching and Teacher Education*, 67, 152–160. <https://doi.org/10.1016/j.tate.2017.06.006>.
- Taylor, J., Banilower, E., & Clayton, G. (2020). National trends in the formal content preparation of US science teachers: Implications of out-of-field teaching for student outcomes. *Journal of Science Teacher Education*, 31(7), 768–779. <https://doi.org/10.1080/1046560X.2020.1762992>.

- United Nations. (2024). *Global report on teachers: Addressing teacher shortages and transforming the profession*. <https://unesdoc.unesco.org/ark:/48223/pf0000388832>
- Vale, C., Campbell, C., Speldewinde, C., & White, P. (2020). Teaching across subject boundaries in STEM: Continuities in beliefs about learning and teaching. *International Journal of Science and Mathematics Education*, 18(3), 463–483. <https://doi.org/10.1007/s10763-019-09983-2>.
- Vale, C., Campbell, C., & White, P. (2021). Beliefs and practices of secondary teachers crossing subject boundaries to teach mathematics out-of-field. *Mathematics Education Research Journal*, 33(3), 589–612. <https://doi.org/10.1007/s13394-020-00323-0>.
- Weldon, P.R. (2016). *Out-of-field teaching in Australian secondary schools*. Policy Insights, Vol. 6. Australian Council for Educational Research (ACER).
- Wittmann, G., Schuler, S., & Levin, A. (2015). To what extent can kindergarten teachers and primary school teachers initiate and foster learning mathematics in typical situations? In K. Krainer & N. Vondrova (Eds.), *Proceedings of the Ninth Conference of the European Society for Research in Mathematics Education (CERME9)* (pp. 3289–3295). Charles Univ, Fac Educ.
- Zhou, Y. (2014). The relationship between school organizational characteristics and reliance on out-of-field teachers in mathematics and science: Cross-national evidence from TALIS 2008. *Asia-Pacific Education Researcher*, 23(3), 483–497. <https://doi.org/10.1007/s40299-013-0123-8>.
- Ziegler, C., & Richter, D. (2017). Effects of out-of-field teaching on student achievement: Can differences in classroom composition explain these effects? [Der Einfluss fachfremden Unterrichts auf die Schülerleistung: Können Unterschiede in der Klassenzusammensetzung zur Erklärung beitragen?]. *Unterrichtswissenschaft*, 45(2), 136–155.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.